LAB SHEET – 09

SMART DIGITAL EXPENSE TRACKER WITH BUDGET PREDICTION

Course: BCA I Semester
Session: 2026–2027
Programming Language: C++

Submitted By: _____

Roll Number: _____

Submitted To: _____

Use this report as a guide and copy it neatly into your practical notebook if a handwritten report is required.

1. Objective

To develop a smart expense management system that helps users record daily expenses, categorize spending, monitor monthly budgets, and predict whether the monthly budget will be exceeded.

2. Problem Statement

Many students spend money daily but do not track their expenses. This project provides an easy way to manage expenses and alerts users before they exceed their monthly budget.

3. Software Requirements

1. C++ programming environment (such as Code::Blocks, Dev-C++ or a C++ compiler).
2. Text editor or IDE.
3. A computer system.

4. Modules / Features

No.	Module	Description
1	User Registration	Creates a new account and stores the username and password.
2	Login System	Checks the entered username and password.
3	Add Daily Expense	Accepts date, category, description and amount.
4	View Expense History	Displays expenses saved by the logged-in user.
5	Category-wise Expense	Calculates spending in Food, Travel, Books, Shopping and Other.
6	Monthly Expense Report	Displays the total expense for a selected month.
7	Budget Setting	Saves or updates a monthly budget.
8	Budget Warning	Warns the user when actual spending is above the budget.
9	Expense Prediction	Estimates month-end spending using average daily spending.
10	Search Expense	Searches by ID, category, description or date.
11	Delete Expense	Deletes a selected expense using its ID.
12	File Saving	Stores user, expense and budget information in text files.

5. Concepts Used

Variables, loops, arrays, structures, functions, file handling, strings, searching, sorting, conditional statements and menu-driven programming.

6. Algorithm / Logic

• Start the program and load previously saved expense records from the file.

• Display the main menu with Registration, Login and Exit options.

• Register a new user or verify the username and password for login.

• After successful login, display the expense tracker menu.

• To add an expense, accept its date, category, description and amount. Calculate the month from the date and save the record.

• Display expense history, category totals or the monthly report when selected.

• Allow the user to set a monthly budget and save it in a file.

• Compare monthly spending with the saved budget and show a warning if the budget is exceeded.

• Estimate month-end spending by multiplying average daily spending by the number of days in the month.

• Allow searching and deleting expenses by ID or search criteria.

• Save changes in files, log out, and stop the program when Exit is selected.

7. Expense Prediction Formula

Predicted Monthly Expense = (Total Expense So Far ÷ Number of Days Elapsed) × Total Days in Month
The prediction is an estimate based on the average daily spending so far. It is not an AI prediction.

8. Files Used

users.txt: Stores registered usernames and passwords.
expenses.txt: Stores expense ID, username, date, month, category, description and amount.
budgets.txt: Stores username, month and budget amount.

9. Sample Input and Output

```
======= SMART DIGITAL EXPENSE TRACKER =======
1. User Registration
2. Login
3. Exit
Enter your choice: 1

--- User Registration ---
Create username: student1
Create password: pass123
Registration successful!

Enter your choice: 2
Username: student1
Password: pass123
Login successful. Welcome, student1!

1. Add Daily Expense
Enter date (YYYY-MM-DD): 2026-11-01
Enter category: Food
Enter short description: Lunch
Enter amount: Rs. 100
Expense added and saved successfully!

Monthly budget: Rs. 3000.00
Total spent: Rs. 100.00
Predicted month-end expense: Rs. 3000.00
```

Note: This is an illustrative sample. The actual prediction depends on the selected month, day number and all expenses entered.

10. Conclusion

The Smart Digital Expense Tracker provides a simple way to record and manage daily expenses. It helps users view spending by category, prepare monthly reports, set budgets, receive budget warnings and estimate future monthly spending. This project demonstrates the practical use of basic C++ programming concepts and file handling.

11. Future Enhancements

AI-based spending prediction, mobile application, QR bill scanner and cloud backup can be added in the future.

12. Important Notes for Demonstration

Use a simple username without spaces. Enter dates in YYYY-MM-DD format and months in YYYY-MM format. Categories should be entered exactly as Food, Travel, Books, Shopping or Other for the category-wise report. Expense descriptions should be one word in this beginner version. The program creates its text data files in the same folder where it is run.

Security note: This is a classroom demonstration. Passwords are stored as plain text, so this design should not be used for real accounts or sensitive information.

Appendix: Source Code

The complete C++ source code is supplied separately as Smart_Digital_Expense_Tracker.cpp. Attach or print that file as the source-code section required by the lab sheet.